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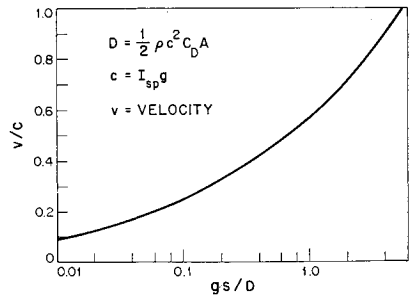


Fig. 1 Control law during thrusting flight.

For $\Omega = 0$, from Eq. (15)

$$\mu = \lambda \left\{ \left[(k-1)v^{k-2} + \frac{v^{k-1}}{c} \right] \frac{\tilde{D}}{m_f} - \frac{g}{v^2} \right\} \qquad (21)$$

The conditions imposed on (15) prior to burnout require that the bracketed term in (21) ≤ 0 for all altitudes. Thus,

$$\mu \leq 0 \qquad y_0 \leq y \leq y_f \qquad (22)$$

In summary, Eqs. (20) and (22) guarantee that the solution is at least a local maximum, with a sufficient requirement that $D = \tilde{D}v^k$, $k > 0$. Other drag laws may indeed be shown to yield valid results, but they are of little interest.

Minimum takeoff weight for given payload to peak altitude specification

The problem of minimum m_0 (or fuel) to achieve a desired altitude may be handled in a similar way. Here the problem is solved backwards from the terminal conditions $s_f = m_f$, $v_f = 0$, y_f-given, and the desired performance index

$$J = \min\{s_0\} \qquad v_0 = 0 \qquad (23)$$

The solution is accomplished by integrating Eq. (4) backward with the preceding end conditions and the appropriate control law. At y_0 the proper conditions on v_0 will require an impulsive jump in the state. For $v_0 = 0$, $m_0 = s_0$. From our previous discussion of sufficiency conditions for a local maximum, it can be shown that for a local minimum (required here), $\mu \geq 0$ and $\partial^2 \bar{H}/\partial v^2 \geq 0$, and that these conditions are met when one integrates the present equation backwards.

Sample Problem

The plant considered by Bryson and Ross[1] was recalculated on a digital computer using the present formulation. This plant assumes constant gravity, an exponential atmosphere, and $D = \tilde{D}v^2$ with a constant drag area. From Eq. (17) the

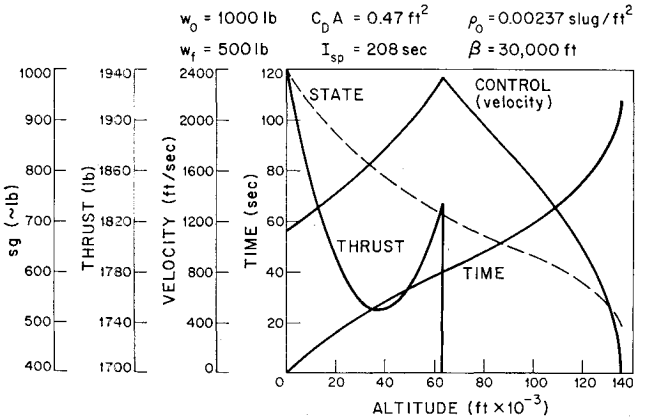


Fig. 2 Sample problem solution.

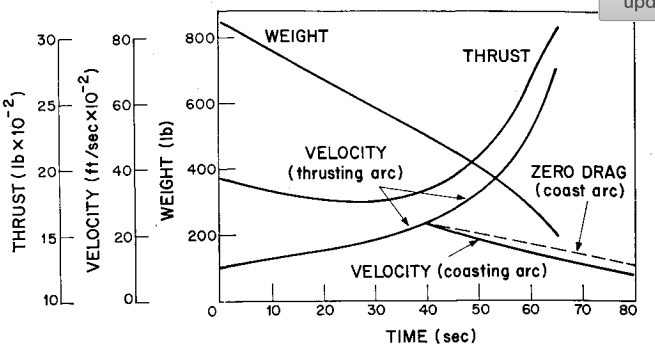


Fig. 3 Sample problem—undetermined burnout weight.

control law becomes

$$\Omega < 0 \qquad \left(1 + \frac{v}{c}\right) e^{v/c} \left(\frac{v}{c}\right)^2 = \frac{gs}{\frac{1}{2}\rho_0 \exp(-y/\beta) c^2 C_D A} \qquad (24)$$

$$\Omega = 0 \qquad s = m_f e^{v/c} \qquad (25)$$

Equation (24) is transcendental in nature and is plotted in Fig. 1. Figure 2 shows profiles of the state, velocity (control), and thrust for a specific set of initial conditions. The results compare favorably with Ref. 1, recognizing that a scale factor error of 2 is present in the figures of that paper. The results show that the state is continuous throughout the three phases of flight.

Since the optimum burning program is affected only by initial conditions prior to burnout, the results for any burnout weight can be easily determined. Figure 3 shows the "sustaining" thrust vs time. For a given burnout weight the thrusting arc is terminated by the appropriate inequality constraint and a corresponding coasting arc is generated as shown for the previous test case ($w_f = 500$ lb). For high altitudes where drag is negligible, the coasting arc has constant slope (for constant gravity) and the peak altitude can be calculated easily.

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Simplified Thermal Fatigue Analysis for Liquid Rocket Combustion Chambers

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Nomenclature

E = modulus of elasticity
K = thermal conductivity

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N_f = number of strain cycles to produce fatigue cracks
T_B = coolant bulk temperature
T_{wc}, T_{wg} = tube coolant wall and tube gas wall temperature, respectively
t = tube wall thickness
α = coefficient of expansion
$\Delta\epsilon_p$ = cyclic plastic strain range
ϵ_e, ϵ_p = elastic and plastic strain, respectively
ϵ_{tot} = total strain
ν_e = Poisson's ratio (elastic)
$\nu_{e-p} = \nu$ = Poisson's ratio (elastic-plastic)
φ = heat flux
σ_{th} = thermal stress (elastic)
σ_y = stress at which plastic deformation first occurs

Subscripts

a = axial, parallel to the tube centerline
eff = effective, or "equivalent" plastic strain as compared to the uniaxial case
r = radial, normal to the tube wall
t = tangential, parallel to a tangent to the tube wall

Introduction

FATIGUE failure in rocket thrust chamber assemblies may be induced through pressure or acceleration force loadings, repeated bending stress loadings, and cyclic temperature variations. In all cases, when fatigue failure occurs after a relatively small number of cycles, the yield point of the material has been exceeded. Both solid and tubular members of regeneratively or mass-transfer-cooled rocket thrust chambers can experience one or more of the foregoing failure initiating mechanisms. In this note attention is focused on low-cycle fatigue cracking in regeneratively cooled tubular-lined thrust chambers (Fig. 1), which is induced by cyclic thermal stress. The failures are typified by the appearance of tangential cracks in the tube wall on the hot-gas side of the tubes. Combustion gases, flowing parallel to the tube axis, heat the exposed tube crowns which are internally cooled, usually by the fuel. Large strains result from the temperature gradients through the hot tube wall and the restraint against axial growth imposed by the colder portions of the tubes and pressure vessel structure. Usually, these axial restraints are of sufficient magnitude to justify the assumption of infinite rigidity in the backup structure. This note presents a means of estimating thrust chamber life or the relative life rating of thrust chamber tube materials in terms of calculable local heat flux and tube temperatures. More rigorous analytical formulations, which require numerical solutions, may then be utilized after the problem has been stabilized. This approach has been used in high-pressure rocket engine studies.[1]

Analysis

A simplified, approximate solution is accomplished by use of the assumptions of "elastic-strain invariance" and isotropic material properties. The first assumption is based on

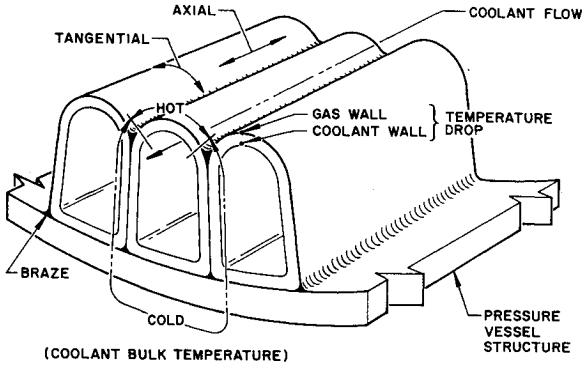


Fig. 1 Model of regeneratively cooled tubular thrust chamber for plastic strain analysis.

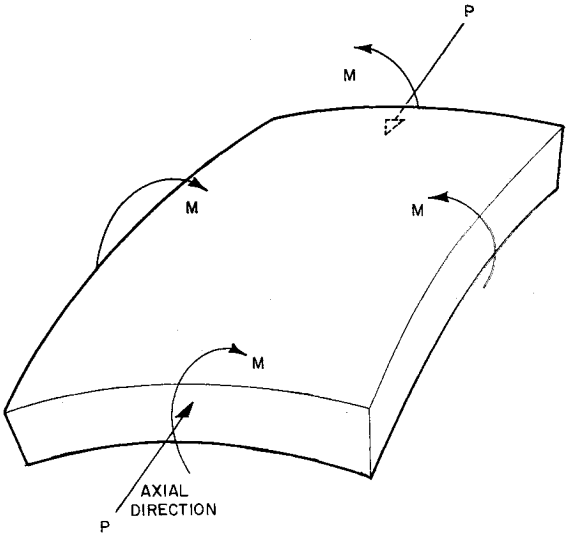


Fig. 2 Flat-plate element free-body diagram.

the observation that the elastically computed strain distribution does not depend on the elastic modulus, if the modulus is constant throughout the body. Elastic theory may then be used to compute the total strains, even if they are partly plastic. If the form of the body and temperature conditions are such that there would be no stresses except for the constraint of external forces, the elastic strains may be found by determining the shape and dimensions the body would assume if unconstrained, and then calculating the strains produced by forcing it to its constrained shape and dimensions. Thus, we consider as a flat-plate free body (Fig. 2), an elemental part that has been removed from the crown of the tube. The temperature difference across the crown,

$$\Delta T_1 = T_{wg} - T_{wc} \tag{1}$$

will cause the element to assume a spherical curvature. The length and width dimensions, as measured along the curvature at the midplane, will increase by an amount proportional to the average temperature rise. If the element was originally at the coolant bulk temperature, the average temperature rise is given by

$$\Delta T_2 = [(T_{wg} + T_{wc})/2] - T_B \tag{2}$$

The constraints against this thermal deformation and growth are applied by the geometry of the thrust chamber tube and jacket. Complete restraint against curvature change in the axial direction is afforded by the curved tube wall. Restraint against growth is realized, for the most part, in the axial direction only. Applying the necessary restraints against curvature to force the element back to its original shape yields the familiar relation

$$\sigma_{th} = \pm E\alpha(\Delta T_1)/2(1 - \nu) \tag{3}$$

Applying the restraints against growth such that the element is forced back to its original dimensions gives $\sigma_{th} = E\alpha(\Delta T_2)$. Substituting $\sigma = E\epsilon$ and superimposing the results of both restraints in the axial direction give, for the tube gas wall,

$$\epsilon_{tot_a} = \alpha(\Delta T_1)/2(1 - \nu) + \alpha(\Delta T_2) \tag{4}$$

Similarly, the restraint against curvature change only in the tangential direction gives

$$\epsilon_{tot_t} = [\alpha(\Delta T_1)/2(1 - \nu)]\beta \tag{5}$$

where β is a factor less than, but usually near, unity for small tube diameters, which accounts for the effects of tube crown curvature and strain relief arising from tangential growth of the hot tube wall. The plastic strain components of the

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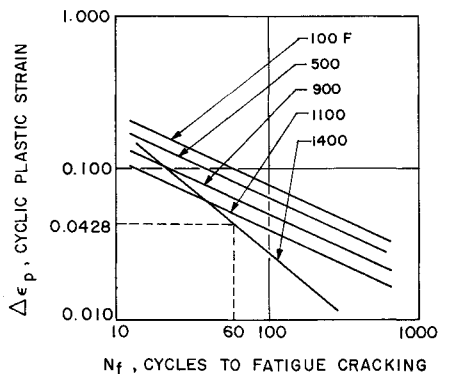


Fig. 3 Low-cycle fatigue diagram, type 347 stainless steel.

total strain may be approximated by

$$\epsilon_p = \epsilon_{tot} - \epsilon_e = \epsilon_{tot} - \sigma_y/E \tag{6}$$

Substituting the previous relationships for ϵ_{tot}, ΔT_1, and ΔT_2 gives for the axial plastic strain at the gas wall:

$$\epsilon_{pa} = \alpha \frac{(T_{wg} - T_{wc})}{2(1-\nu)} + \alpha \left(\frac{T_{wg} + T_{wc}}{2} - T_B \right) - \frac{\sigma_y}{E} \tag{7}$$

whereas for the tangential plastic strain,

$$\epsilon_{pt} = \alpha\beta(T_{wg} - T_{wc})/2(1-\nu) - \sigma_y/E \tag{8}$$

The foregoing relationships, while only approximate for plastic strain computations, are particularly useful for parametric studies on the heat transfer and plastic strain interrelations. The limiting heat-transfer conditions for zero plastic strain may be determined from Eq. (7) by rearranging to the form

$$T_{wg} = T_B - \left(\frac{T_{wg} - T_{wc}}{2} \right) \left(\frac{\nu_e}{1-\nu_e} \right) + \frac{\sigma_y}{E\alpha} \tag{9}$$

and substituting the conductive heat-transfer relationship

$$T_{wg} - T_{wc} = \varphi t/K \tag{10}$$

giving

$$T_{wg} = T_B - (\varphi t/2K)[\nu_e/(1-\nu_e)] + \sigma_y/E\alpha \tag{11}$$

Here, T_{wg} is the maximum gas wall temperature, which may be sustained while operating at a heat flux φ in order not to exceed the yield stress σ_y. Except for very low heat fluxes, almost all regenerative tube designs operate with the tube in a plastic straining condition. For example, a 0.012-in.-thick type 347 annealed stainless-steel tube wall operating at a heat flux of $\varphi = 2$ Btu/in.^{2}/sec, and a coolant bulk temperature of 100°F, may not exceed 194°F on the gas wall, if plastic strain is to be avoided.

From the theory of plasticity, it is known that the maximum or "effective" plastic strain at the tube gas wall is greater than either the axial or tangential plastic strain individually. To apply uniaxial cyclic fatigue test data to the prediction of tube fatigue life, a relationship between the uniaxial test data and the real case is used. This relationship may be taken from the "distortion energy" theory of plasticity. Accordingly, the approximate effective plastic strain is given by (see Ref. 2 for a discussion of this theory)

$$\epsilon_{peff} = \frac{(2)^{1/2}}{3} [(\epsilon_{pa} - \epsilon_{pt})^2 + (\epsilon_{pt} - \epsilon_{pr})^2 + (\epsilon_{pr} - \epsilon_{pa})^2]^{1/2} \tag{12}$$

This may be simplified by the introduction of the "no volume change" basic assumption of all plasticity theory. Then, for fully developed isotropic plastic flow,

$$\epsilon_{pr} = -\epsilon_{pa} - \epsilon_{pt} \tag{13}$$

Substituting this result in Eq. (12) gives

$$\epsilon_{peff} = \frac{2}{(3)^{1/2}} (\epsilon_{pt}^2 + \epsilon_{pt}\epsilon_{pa} + \epsilon_{pa}^2)^{1/2} \tag{14}$$

If the total plastic strain occurring during one engine run is that incurred from the start and shutdown temperature cycle, the cyclic plastic strain per engine run is $\Delta\epsilon_p = 2\epsilon_{peff}$, assuming that the stress-strain hysteresis loop returns to its original starting point. In reality, this occurs after several run cycles, when the tube material is fully strain hardened. Also, plastic strain cycling may occur during steady-state run conditions, if the induced "steady-state" cyclic thermal stresses exceed the yield point of the material.

The calculated $\Delta\epsilon_p$ may then be entered in low-cycle fatigue life diagrams for the particular tube material being studied. The life may be determined directly from the diagram, or if computer usage is desired, from the low-cycle fatigue relationship $N_f{}^k \Delta\epsilon_p = C$, where the constant C and exponent k are known functions of temperature. The life thus determined is given in terms of total engine runs to fatigue cracking of the tubes.

Example

The following example typifies the expected life of a hydrogen-cooled type 347 stainless-steel thrust chamber tube:

$$T_{wg} = 1400°\text{F}\ T_{wc} = 1060°\text{F}\ T_B = -250°\text{F}$$

$$\varphi = 9\ \text{Btu/in.}^2\text{-sec}\ \sigma_y = 20\ \text{ksi}\ E = 20 \times 10^4\ \text{psi}$$

$$\alpha = 10 \times 10^{-6}\ \text{in./in.-°F}\ \nu = 0.5$$

Then

$$\epsilon_{pa} = 0.01720\ \epsilon_{pt} = 0.00240$$

$$\epsilon_{peff} = 0.0214,\ \Delta\epsilon_p = 0.0428$$

Entering a stainless-steel 347 low-cycle fatigue diagram (Fig. 3) with this value, at a wall temperature of 1400°F, shows that there would be $N_f = 60$ engine runs to fatigue cracking.

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